problems/MHD/smoothVortex

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. Read off `user_inputs.jl`, the case files and the kernels they reach on `sm/MHD`; deck values are those obtained with every `JEXPRESSO_SV_*` override unset. The case is the smooth-flow accuracy test (Dao & Nazarov 2022, §5.1.1; Balsara 2004): its point is that the residual viscosity must *not* degrade a smooth high-order solution, so it is run against a plain-Galerkin twin (`JEXPRESSO_SV_VISC=none` sets `:lvisc => false`).

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	2D ideal GLM-MHD, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$	`initialize.jl`; flux code identical to Orszag–Tang's
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2 - \frac{1}{2}\psi^2)$, $\gamma = 5/3$	`user_flux.jl`
Domain, mesh	$[-10, 10]^2$ ($L = 20$), doubly periodic, 16×16 quads	`vortex_L20_16x16.msh` (`JEXPRESSO_SV_L`, `_NELX`)
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 4$	`JEXPRESSO_SV_NOP`
Time integration	`CarpenterKennedy2N54`, fixed Δt, $\Delta t = 2 \times 10^{-3}$, $t_{end} = 1$	$\Delta t = 2 \times 10^{-3} \cdot 64/(n_{el}N)$; `JEXPRESSO_SV_SOLVER` can switch to Vern9, DP8, ...
Variables	`TOTAL()`, `"energy"`	
Initial data	$\rho = 1$, $\mathbf{v} = (1,1) + \kappa g\,(-y, x)$, $\mathbf{B} = \mu g\,(-y, x)$, $g = e^{(1-r^2)/2}/(2\pi)$, $p = 1 + \frac{\mu^2(1-r^2)-\kappa^2}{8\pi^2} e^{1-r^2}$, $\kappa = \mu = 1$	`sv_state`
GLM cleaning	$c_h = \max(\lvert\mathbf{v}\rvert + c_f)$ of the initial data, constant; source $S_\psi = -(c_h/0.18)\,\psi$	`initialize.jl`, `user_source.jl`
Sources	`:lsource => true` (the ψ damping only)	
Boundary conditions	periodic (the free-slip fallback of `user_bc.jl` never fires)	
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false	

2 DynSGS: deck keys and effective values

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`, `:lvisc => true`	2D element kernel `SGS.jl:1598`
`:`μ	all 1	

Deck key / item	Value as run	Source / remark
`:dsgs_sensor`	`"residual"`	`JEXPRESSO_SV_SENSOR`
`:dsgs_CR`	1.0	`JEXPRESSO_SV_CR`
`:dsgs_Cmax`	0.5	`JEXPRESSO_SV_CMAX`
`:dsgs_Cmin`	0	`JEXPRESSO_SV_CMIN`
`:dsgs_gamma`	$= \gamma_{mhd} = 5/3$	
`:dsgs_Prt`	1.0	$\kappa = \bar{\rho}\nu/\,\mathrm{Pr}$
`:dsgs_rel`	1.0	`JEXPRESSO_SV_REL`
`:dsgs_norms`	`"domain"`	`JEXPRESSO_SV_NORMS`
`:dsgs_conserved`	**false**	physical form, since `f9cdb02` (was `true`: Section 4)
`:dsgs_nazarov_energy`	**true**	since `f9cdb02`; $\kappa = \bar{\rho}\nu/\,\mathrm{Pr}$ on ∇T
`:dsgs_fast_floors`, `:dsgs_thermal_residual`	false (defaults)	
`:dsgs_ref_weight`, `:dsgs_nodal_rho`, `:dsgs_reference`	false, false, false	defaults
`:dsgs_hold_steps`	2	`JEXPRESSO_SV_HOLD`
`:dsgs_cutoff`	0	`JEXPRESSO_SV_CUTOFF`
`:dsgs_freeze_stage`	false	`JEXPRESSO_SV_FREEZE`
`:ldsgs_nodal`	false (element form)	default
`:dsgs_local_rel`	1.0 (default)	unused (domain norms)

3 DynSGS as it runs for this case

Element K, node $I = (i, j)$, $m_I^K = \omega_i \omega_j J_{K,ij}$, $\Delta_K = \Delta_K^{elem}/(N+1)$ with Δ_K^{elem} the smaller corner-to-corner extent of K: $\Delta_K^{elem} = 1.25$, $\Delta_K = 0.25$.

1. **History and hold.** Rotation once per step; two held steps ($\nu = 0$ then, since $C_{\min} = 0$).
2. **Time derivative.** Residual sensor: BDF2 on $(\mathbf{q}^n, \mathbf{q}^{n-1}, \mathbf{q}^{n-2})$ at the stage on t^n, the three-point derivative through $(\mathbf{q}(\tau), \mathbf{q}^n, \mathbf{q}^{n-1})$ at later stages (formulas as in `rhs.jl:964-971`).
3. **Residual.** $R_{K,I}^s = |D_t q_I^s - r_I^{K,s}/m_I^K|$, r^K the element's own weak inviscid RHS (flux divergence and the ψ source). The mesh is periodic, so the Dirichlet-node list is empty and nothing is zeroed.
4. **Normalization** (domain, MPI-reduced): $n^s = \max(S^s, \sigma^s) + 10^{-16}$ with the mean-state floors $\sigma^\rho = \bar{\rho}$, $\sigma^{\mathbf{m}} = \bar{\rho}\bar{c}$, $\sigma^E = \bar{\rho}\bar{c}^2$, $\sigma^{\mathbf{B}} = \sqrt{\bar{\rho}}\,\bar{c}$, $\bar{c} = \sqrt{\gamma\bar{p}/\bar{\rho}}$. Here $\rho \equiv 1$, so $S^\rho = 0$ and the continuity residual is measured against $\bar{\rho} = 1$ (this is the case for which the floor was raised from 10^{-3} of the scale to the scale itself, `:dsgs_rel`).
5. **Ratio and cap.** $\mathcal{R}_K = \max_{I \in K} \max_{s \le 8} R_{K,I}^s/n^s$ (ψ excluded); $w_K = \max_I(|\mathbf{v}_I| + \sqrt{\max(\gamma p_I/\rho_I + |\mathbf{B}_I|^2/\rho_I, 0)})$.
6. **Coefficient.** $\nu_K = \max(0, \min(0.5\,\Delta_K w_K,\ \Delta_K^2 \mathcal{R}_K))$, no floor, no cutoff, one value per element, recomputed at every stage.
7. **Slot coefficients** (physical branch, `SGS.jl:1889-1900`), with $\bar{\rho}_K$ the plain average of the nodal ρ over the 25 nodes of K: $\mu^\rho = \nu_K$, $\mu^{\rho u} = \mu^{\rho v} = \mu^{\rho w} = \bar{\rho}_K \nu_K$, $\mu^E = \bar{\rho}_K \nu_K/\,\mathrm{Pr} = \bar{\rho}_K \nu_K$, $\mu^{\mathbf{B}} = \mu^\psi = \nu_K$.

4 The viscous operator

`user_primitives.jl` returns $(\rho, u, v, T{=}p/\rho, w, B_x, B_y, B_z, \psi)$ and, with `lcons = false`, `_expansion_visc!` (`rhs.jl:2943`) assembles the same physical form as Orszag–Tang:

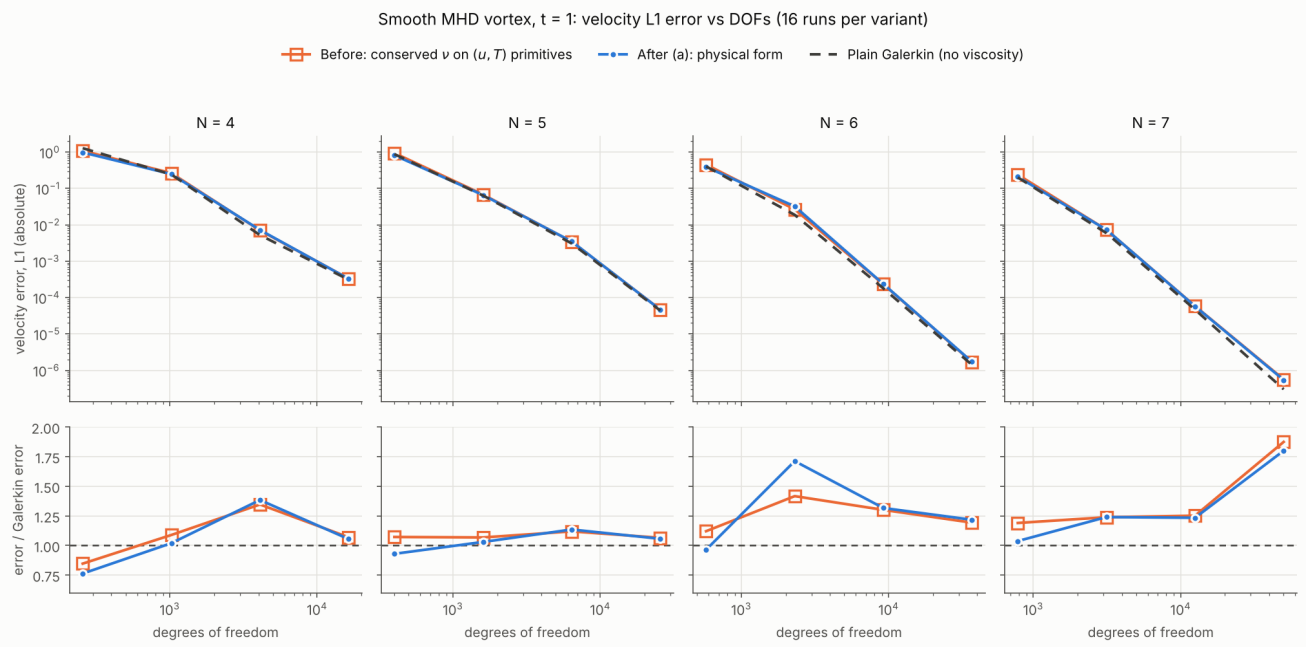
Equation	Viscous flux $\mathbf{F}_V$	Coefficient
ρ	$\nu\nabla\rho$	mass diffusion on
$\rho u,\ \rho v$	$\boldsymbol{\tau} = \bar{\rho}\nu\big(\nabla\mathbf{v} + \nabla\mathbf{v}^T - \frac{2}{3}(\nabla\cdot\mathbf{v})I\big)$, $\mathbf{v} = (u, v)$	$\bar{\rho}\nu$
ρw	$\bar{\rho}\nu\nabla w$	$\bar{\rho}\nu$
E	$\bar{\rho}\nu\nabla T + \boldsymbol{\tau}\cdot\mathbf{v} + \nu\sum_k B_k\nabla B_k$	heat + viscous work + resistive work
B_k	$\nu\nabla B_k$	$\eta = \nu$
ψ	$\nu\nabla\psi$	

What it replaced (finding 2 of `summary.pdf`, fixed in `f9cdb02`)

At `dcc3653` the deck had `:dsgs_conserved => true` with these same physical primitives. The kernel then gave every slot the kinematic ν_K and `_expansion_visc!` dropped the stress and the work terms (`rhs.jl:2959-2970`): $\nu\nabla u$, $\nu\nabla v$, $\nu\nabla w$ on the momenta, $\nu\nabla T$ on E — neither the conserved form (which needs $\nabla(\rho\mathbf{v})$, ∇E) nor the physical one, and the kinetic and magnetic energy removed by those terms was not returned to E. The header of `user_primitives.jl` already described the physical form.

Test: before against after

Run here with Julia 1.11.5 on the deck as committed, with every `JEXPRESSO_SV_*` override unset except N and the mesh: the grid of `tools/smooth_vortex_scan.sh`, $N = 4, \ldots, 7$ on 4^2, 8^2, 16^2, 32^2 elements of the $L = 20$ box, $\Delta t = 2\times10^{-3}\cdot 64/(n_{el}N)$, `CarpenterKennedy2N54`, $t = 1$; 16 runs for each of three variants: *before* (the deck of `dcc3653`), *after* (`f9cdb02`) and the plain Galerkin scheme (`JEXPRESSO_SV_VISC=none`, which ignores every DynSGS key). All 48 runs completed. The error is the case's own absolute velocity error against the exact solution (`user_plot.jl`); orders are against $h \propto \mathrm{DOFs}^{-1/2}$ between consecutive meshes.



N	mesh	DOFs	velocity L^1 error			after /before	before /Gal.	after /Gal.	observed order		
			before	after (a)	Galerkin				before	after	Gal.
4	4^2	256	1.09×10^{0}	9.82×10^{-1}	1.29×10^{0}	0.901	0.85	0.76			
4	8^2	1024	2.65×10^{-1}	2.48×10^{-1}	2.44×10^{-1}	0.936	1.09	1.02	2.04	1.98	2.40
4	16^2	4096	7.05×10^{-3}	7.25×10^{-3}	5.24×10^{-3}	1.028	1.35	1.38	5.23	5.10	5.54
4	32^2	16384	3.33×10^{-4}	3.28×10^{-4}	3.11×10^{-4}	0.986	1.07	1.05	4.40	4.46	4.07
5	4^2	400	9.45×10^{-1}	8.20×10^{-1}	8.82×10^{-1}	0.867	1.07	0.93			
5	8^2	1600	6.86×10^{-2}	6.61×10^{-2}	6.42×10^{-2}	0.964	1.07	1.03	3.78	3.63	3.78
5	16^2	6400	3.46×10^{-3}	3.51×10^{-3}	3.09×10^{-3}	1.014	1.12	1.14	4.31	4.24	4.38

N	mesh	DOFs	velocity L^1 error			after	before	after	observed order		
			before	after (a)	Galerkin	/before	/Gal.	/Gal.	before	after	Gal.
5	32^2	25600	4.70×10^{-5}	4.66×10^{-5}	4.41×10^{-5}	0.992	1.07	1.06	6.20	6.23	6.13
6	4^2	576	4.55×10^{-1}	3.91×10^{-1}	4.06×10^{-1}	0.861	1.12	0.96			
6	8^2	2304	2.67×10^{-2}	3.23×10^{-2}	1.89×10^{-2}	1.209	1.42	1.71	4.09	3.60	4.43
6	16^2	9216	2.34×10^{-4}	2.37×10^{-4}	1.80×10^{-4}	1.012	1.30	1.32	6.84	7.09	6.72
6	32^2	36864	1.68×10^{-6}	1.72×10^{-6}	1.41×10^{-6}	1.020	1.19	1.22	7.12	7.11	6.99
7	4^2	784	2.40×10^{-1}	2.10×10^{-1}	2.02×10^{-1}	0.872	1.19	1.04			
7	8^2	3136	7.36×10^{-3}	7.37×10^{-3}	5.94×10^{-3}	1.001	1.24	1.24	5.03	4.83	5.09
7	16^2	12544	5.85×10^{-5}	5.77×10^{-5}	4.68×10^{-5}	0.986	1.25	1.23	6.97	7.00	6.99
7	32^2	50176	5.66×10^{-7}	5.43×10^{-7}	3.02×10^{-7}	0.960	1.87	1.80	6.69	6.73	7.28

Reading. On the two finest meshes the two operators are within -4% and $+3\%$ of each other in L^1 and their observed orders agree to ± 0.25 (finest pair: 4.40/4.46, 6.20/6.23, 7.12/7.11, 6.69/6.73 for $N = 4, \ldots, 7$); on the coarsest (4^2) the physical form is 10–14% lower in L^1 and 7–28% lower in L^∞; the one point where it is clearly worse is $N = 6$ on 8^2 ($1.21\times$ in L^1). Both stay within a factor of 1.05–1.4 of the plain Galerkin error at $N \leq 6$ on the finest meshes, and both are ≈ 1.8–$1.9\times$ Galerkin at $N = 7$ on 32^2, where the viscosity itself, common to both, sets the gap. The change of operator therefore does not alter the accuracy picture of this case; it makes the operator the one the case documents.

5 Positivity

- `:lpositivity` false; no stage limiter; no state modification of any kind.
- No flux-only floor either: `pressure_mhd` is the unguarded Orszag–Tang one.
- None is needed by design: the data are smooth, $\min p = 1 - 1/(8\pi^2) \approx 0.987$ initially (at $r = 1$), and the case measures accuracy, not robustness.

6 Observations: code versus comments

1. Since `f9cdb02` the deck, `user_primitives.jl` and the kernel agree (the physical form above).
2. Everything else in the deck matches its comments (domain norms, hold 2, no floor, no cutoff, the residual sensor).

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	**Where (file:line)**
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report	`src/io/mod_inputs.jl:456-477`
`:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H) + 1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\,\mathbf{B} \cdot \nabla\mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`

Mechanism	**Where (file:line)**
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs}+2$ in 2D, $n_{eqs}+1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)